

Electric Potential Energy (completed)

Goal.

- Understand where electric potential energy comes from: summing over pairs of point charges, or pushing a charge against a field.
- Use that to find the potential energy of an electric dipole in a uniform field.
- See that a system can have *negative* PE, and that “zero” PE is not always the lowest energy available.

1. Potential energy of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy.

$$U(q_1, q_2) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

- Keep the **signs** of the charges — U can come out positive *or* negative.
- As $r_{12} \rightarrow \infty$, $U \rightarrow 0$.
- For three or more charges, sum over every **pair** (not every charge).
- Opposite charges (+ and -): U is **negative** at any finite separation, and $U \rightarrow 0$ only as $r \rightarrow \infty$. So $U = 0$ is actually the **largest** value here — pulling the charges closer always finds a *lower* (more negative) PE.
- Like charges (same sign): U is **positive** at any finite separation, and again $U \rightarrow 0$ as $r \rightarrow \infty$. Here $U = 0$ is the **smallest** value — you can’t push like charges to a lower PE than infinitely far apart.

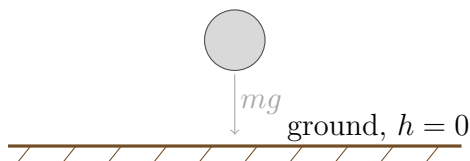
$$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$$

Check. Two $+q$ charges are brought from far apart to a separation r . Is U positive or negative? What about a $+q$ and a $-q$?

$+q$ and $+q$: positive (repulsive, energy stored pushing them together). $+q$ and $-q$: negative (attractive).

2. Where is potential energy the lowest?

Draw a ball, and the ground below it.



Check. What is the ball's potential energy? Where is it the lowest?

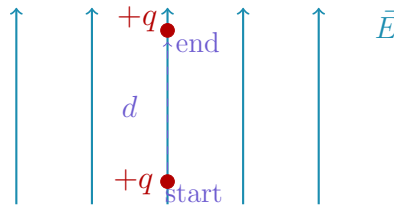
$U = mgh$; with h measured from the ground, $U = 0$ there. As drawn, the ball can't go any lower than the ground — so here $h = 0$ really is the lowest available PE (unlike the attractive point charges above, there's no “digging deeper” available).

This is a *uniform* gravitational field — not a hole to fall into. Raising the ball, moving it against the field lines, gives it *more* PE, and that PE is proportional to the distance you raise it.

$$\Delta U_{\text{grav}} = mgd$$

3. The same idea, for a charge in a uniform field

Now replace the ball with a positive charge q , and gravity with a uniform electric field $\vec{E}$. Draw the field, and the charge's starting and ending points, below.



The field pushes the charge with force qE . Push the charge a distance d against the field — the same motion as raising the ball.

$$\Delta U = qEd$$

- This is exactly like raising the ball: you do **positive** work pushing against the field, and that work becomes stored PE, proportional to **the distance d** .
- Let the charge go, and it moves *with* the field instead: now ΔU is **negative** — the field does the work, and spends the stored PE.

Check. A positive charge is released and moves along $\vec{E}$. Does its PE increase or decrease? Where did the energy go?

PE decreases — the charge is moving *with* the field, the way the ball falls rather than rises. The lost PE becomes kinetic energy.

4. The dipole in a field

A dipole is a $+q$ and a $-q$ separated by $\vec{d}$ (dipole moment $\vec{p} = q\vec{d}$), sitting in a uniform field $\vec{E}$, tilted at angle θ between $\vec{p}$ and $\vec{E}$. Each charge sits at a different place along $\vec{E}$ — use §3's pushing idea on each charge, then add.

Split $\vec{d}$ into a component along $\vec{E}$ ($d \cos \theta$) and a component across it. Only the along- $\vec{E}$ part matters (§3): the $+q$ sits $\frac{1}{2}d \cos \theta$ further along $\vec{E}$ than center, the $-q$ sits $\frac{1}{2}d \cos \theta$ behind it. Adding the two charges' PE (each $\propto -qEd$, from §3, since moving *with* $\vec{E}$ lowers PE) gives $U = -qEd \cos \theta = -\vec{p} \cdot \vec{E}$.

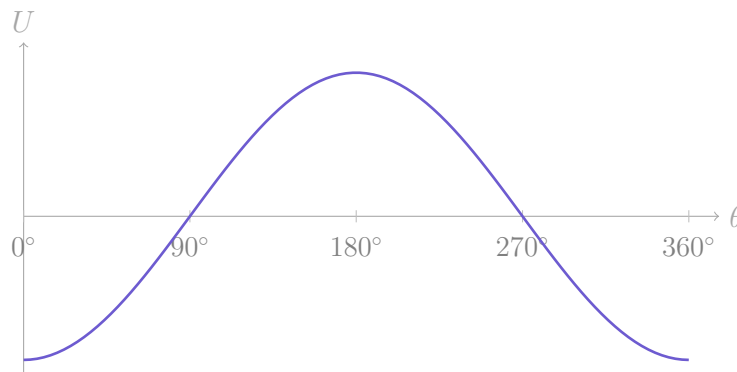
$$U(\theta) = -pE \cos \theta$$

- At $\theta = 90^\circ$ (perpendicular to the field): $U = 0$.
- At $\theta = 0^\circ$ (aligned with the field): $U = -pE$ — which is **lower** than the $\theta = 90^\circ$ value!
- At $\theta = 180^\circ$ (anti-aligned): $U = +pE$.

Check. Which orientation is the most stable (lowest PE)? Is that the orientation where $U = 0$? What does this have in common with the ball in the gravitational field?

$\theta = 0^\circ$ (aligned) is most stable, at $U = -pE$ — *not* the $U = 0$ orientation. Just like the ball: $U = 0$ is just wherever we started measuring from, not necessarily the lowest energy the system can reach.

5. Sketch it: U vs. θ



Check your curve, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.